

Author Technical Correction and Updated Analysis

For ‘Quantum computation capability verification protocol for noisy intermediate-scale quantum devices with the dihedral coset problem’

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Status. This is an author-maintained technical correction hosted with the code repository. It is not an APS Erratum, has not been peer reviewed as a replacement article, and does not alter the journal version. The statements here are made by Ruge Lin and should not be read as a statement on behalf of the journal or as an assertion of endorsement by the coauthor.

Abstract

The 2022 article proposed a DCP-derived circuit, called ParitySolve, and interpreted its success probability relative to a quantity denoted by p_B as verification of quantum-computation capability. A later reanalysis shows that the circuit-level parity extraction is correct, but the verification interpretation is not. The quantity p_B is the success probability of one specified all-Hadamard decoder; it is neither proved nor valid as an upper bound over all strategies using the same product measurements and classical postprocessing. An exact two-sample counterexample gives success 25/32, compared with 23/32 for the published decoder, without adding any quantum operation. Consequently, the acceptance condition $p > p_B$ does not establish the claimed capability.

The reanalysis also strengthens the fair part of the original baseline: for one all-Hadamard sample, the published special-outcome rule is Bayes-optimal among all classical decoders of the complete measurement record. Its failure begins only when several samples reuse the same hidden secret and jointly reveal correlations. This note records what survives, proves the one-sample theorem, supplies an exact formula for the ideal ParitySolve probability, corrects the Figure 5 parameter interpretation, and replaces the reported fluctuation formula by the Bernoulli standard error. The defensible surviving classification is a structured DCP-derived circuit workload, not an adversarially sound verification protocol.

1. Scope and bibliographic record

This note concerns:

Ruge Lin and Weiqiang Wen, “Quantum computation capability verification protocol for noisy intermediate-scale quantum devices with the dihedral coset problem,” *Physical Review A* **106**, 012430 (2022), DOI: 10.1103/PhysRevA.106.012430, arXiv:2202.06984.

The original repository state is preserved by the immutable GitHub release and tag `paper-2022-original`. The corrected repository adds new files but leaves the six original Python scripts unchanged.

The correction has four purposes:

1. preserve the valid circuit construction;
2. withdraw the unsound interpretation of p_B ;
3. replace approximate or unsupported analytical statements by exact reproducible results;
4. separate a later DCP-derived witness from the claims of the 2022 article.

The four-qubit IBM run was a proof-of-concept experiment. It remains in the historical record and is outside the scope of this correction.

2. Circuit result that remains correct

Let $N = 2^n$, and let a DCP sample be

$$|\psi_{x,s}\rangle = \frac{|0\rangle|x\rangle + |1\rangle|x+s \bmod N\rangle}{\sqrt{2}}, \quad x, s \in \mathbb{Z}_N.$$

After applying the Fourier transform F_N to the rotation register and measuring the Fourier label k , the reflection qubit is, up to a global phase,

$$|\phi_{k,s}\rangle = \frac{|0\rangle + \omega_N^{ks}|1\rangle}{\sqrt{2}}, \quad \omega_N = e^{2\pi i/N}.$$

Suppose two measured labels satisfy

$$k_1 - k_2 \equiv \frac{N}{2} \pmod{N}.$$

On the selected output branch of the two-reflection-qubit CNOT operation, the remaining relative phase is

$$\omega_N^{(k_1-k_2)s} = \omega_N^{Ns/2} = (-1)^s.$$

A final Hadamard measurement therefore returns $s \bmod 2$ exactly in the ideal model. This ParitySolve algebra is correct. The circuit remains a meaningful workload involving DCP preparation, Fourier transforms, measurement, collision search, inter-cell interference, and readout.

The correction does not withdraw this circuit identity.

3. The original baseline and its limitation

3.1 What p_B actually computes

The article considered applying Hadamard gates to every qubit of every DCP sample. Write one measurement outcome as (r, y) , where $r \in \{0, 1\}$ is the reflection result and $y \in \mathbb{Z}_N$ is the rotation-register result.

After averaging over the uniform preparation label x , the exact conditional probability is

$$q_s(r, y) = \frac{1}{N^2} \# \{x \in \mathbb{Z}_N : r = x \cdot y \oplus ((x + s) \bmod N) \cdot y\},$$

where the dot denotes the binary inner product.

For $y = 1$, only the least significant bit is selected, so

$$r = x_0 \oplus (x + s)_0 = s \bmod 2.$$

Thus one sample yields a special parity-revealing outcome with probability $1/N$. The published decoder retained only this outcome and guessed uniformly if none of the mt samples produced it. Its success probability is

$$p_B = 1 - \frac{1}{2} \left(\frac{N-1}{N} \right)^{mt}.$$

This formula is correct for that specific decoder. It should be read as the *special-outcome all-H decoder probability*, not as an optimized soundness bound.

3.2 Exact one-sample optimality theorem

The one-sample baseline is stronger than the original article proved.

Let $P_b(r, y)$ denote the outcome distribution averaged uniformly over secrets of parity $b \in \{0, 1\}$. Then

$$P_0(r, y) - P_1(r, y) = \frac{(-1)^r}{N} \mathbf{1}_{\{y=1\}}.$$

Therefore the complete one-sample all-H record contains parity information only when $y = 1$. The equal-prior Bayes-optimal success probability is

$$\frac{1}{2} + \frac{1}{2N},$$

which is exactly the success of the published one-sample special-outcome decoder.

Proof

For fixed x and s , the all-H outcome probability is

$$p(r, y \mid x, s) = \frac{1}{2N} \left[1 + (-1)^{r+x \cdot y + ((x+s) \bmod N) \cdot y} \right].$$

Averaging over uniform x gives $q_s(r, y)$. Taking the even-parity average minus the odd-parity average cancels the constant term and yields

$$P_0(r, y) - P_1(r, y) = \frac{(-1)^r}{N^3} \sum_{x,s} (-1)^{s+x \cdot y + ((x+s) \bmod N) \cdot y}.$$

Set

$$z = (x + s) \bmod N.$$

The map $(x, s) \mapsto (x, z)$ is a bijection. Since N is even,

$$s \bmod 2 = x_0 \oplus z_0.$$

The sum therefore factorizes:

$$P_0(r, y) - P_1(r, y) = \frac{(-1)^r}{N^3} \left(\sum_x (-1)^{x_0 + x \cdot y} \right) \left(\sum_z (-1)^{z_0 + z \cdot y} \right).$$

Each Walsh sum equals N precisely when y is the least-significant-bit vector, identified with the integer 1, and equals zero otherwise. This proves the displayed identity.

The total-variation distance is consequently

$$\text{TV}(P_0, P_1) = \frac{1}{2} \sum_{r,y} |P_0(r, y) - P_1(r, y)| = \frac{1}{N}.$$

For equal priors, Bayes success is

$$\frac{1}{2} (1 + \text{TV}(P_0, P_1)) = \frac{1}{2} + \frac{1}{2N}.$$

The published decoder attains this value by answering when $y = 1$ and guessing otherwise. Hence it is optimal for one all-Hadamard outcome. $\square$

3.3 Why several samples are different

The problem is that all mt samples in one repetition share the same hidden secret s . The relevant parity-conditioned record is

$$\frac{2}{N} \sum_{s \equiv b \pmod{2}} q_s^{\otimes mt},$$

not the tensor power of the one-sample parity average. Outcomes that are parity-neutral individually can therefore be correlated through the reused secret and become informative when processed together.

The original baseline was not unreasonable at the one-sample level. It was incomplete at the shared-secret multi-sample level.

3.4 Exact two-sample counterexample

Take

$$N = 4, \quad m = 2, \quad t = 1.$$

Use exactly the same all-H product measurements as the published baseline. The published decoder succeeds with

$$p_B = 1 - \frac{1}{2} \left(\frac{3}{4} \right)^2 = \frac{23}{32}.$$

Now use the following classical decoder on the same two outcomes (r_1, y_1) and (r_2, y_2) :

1. if either $y_j = 1$, return the corresponding r_j ;
2. otherwise, if $y_1, y_2 \in \{2, 3\}$, return $r_1 \oplus r_2$;
3. otherwise, guess uniformly.

The first event occurs with probability

$$1 - \left(\frac{3}{4} \right)^2 = \frac{7}{16}$$

and reveals the parity exactly.

The additional event $y_1, y_2 \in \{2, 3\}$ occurs with probability $1/4$, is disjoint from the first event, and gives conditional success $3/4$: for even s , the XOR is always zero; for odd s , it is uniform. Hence

$$\begin{aligned} p_{\text{better}} &= \frac{7}{16} + \frac{1}{4} \cdot \frac{3}{4} + \left(1 - \frac{7}{16} - \frac{1}{4} \right) \frac{1}{2} \\ &= \frac{25}{32}. \end{aligned}$$

The improvement is

$$\frac{25}{32} - \frac{23}{32} = \frac{1}{16}.$$

No entangling operation, quantum memory, alternative basis, additional sample, or quantum postprocessing is used. The improvement comes solely from retaining correlations discarded by the original classical decoder.

This disproves the interpretation of p_B as a general upper bound for the intended weaker class. Therefore

$$p > p_B \implies \text{verified quantum-computation capability}$$

is withdrawn.

3.5 Full likelihood decoder

For an all-H record $o_1, \dots, o_\ell$, with $\ell = mt$, the Bayes-optimal parity likelihoods are

$$L_b(o_1, \dots, o_\ell) = \frac{2}{N} \sum_{\substack{s=0 \\ s \equiv b \pmod{2}}}^{N-1} \prod_{j=1}^{\ell} q_s(o_j), \quad b \in \{0, 1\}.$$

Returning the larger likelihood uses all information in the original measurement record. The repository implements this decoder exactly for small cases and by fixed-seed Monte Carlo for the larger published instances.

4. Exact ideal ParitySolve probability

The original analysis used upper and lower bounds for the probability of finding a complementary Fourier-label collision. An exact finite formula is available.

Partition the N labels into $N/2$ complementary pairs

$$\{a, a + N/2\}.$$

Let $k_{\text{nc}}(N, m)$ be the probability that m independent uniform labels contain no complementary pair. If exactly j complementary pairs are occupied without a collision, then:

1. choose the occupied pairs: $\binom{N/2}{j}$;
2. choose one member of each pair: 2^j ;
3. assign the m labelled samples surjectively to the j selected labels: $j! \left\{ \begin{smallmatrix} m \\ j \end{smallmatrix} \right\}$.

Therefore

$$k_{\text{nc}}(N, m) = \frac{1}{N^m} \sum_{j=1}^{\min(m, N/2)} \binom{N/2}{j} 2^j j! \left\{ \begin{smallmatrix} m \\ j \end{smallmatrix} \right\},$$

where $\left\{ \begin{smallmatrix} m \\ j \end{smallmatrix} \right\}$ is a Stirling number of the second kind. The case $m = 0$ is defined as $k_{\text{nc}} = 1$.

In one batch, a complementary collision exists with probability $1 - k_{\text{nc}}$. The selected CNOT branch succeeds with probability $1/2$. Under the retry rule implemented in the original code, one batch fails to return a definitive parity with probability

$$\frac{1 + k_{\text{nc}}}{2}.$$

After t batches, the exact ideal success probability, including a final random guess, is

$$p_{\text{exact}} = 1 - \frac{1}{2} \left(\frac{1 + k_{\text{nc}}}{2} \right)^t.$$

The repository checks the formula against exhaustive enumeration for all test instances with $n \leq 3$ and $m \leq 4$.

5. Corrected interpretation of Figure 5

The original parameter search selected instances from the difference between the published upper bound p_{upper} and p_B . That is a legitimate bound comparison, but it does not establish the actual honest separation.

Figure	(n, m, t)	p_B	Published p_{upper}	Exact honest p	Full-likelihood all-H
5(a)	(4, 6, 1)	0.660533	0.664083	0.652965	0.810940 , exact
5(b)	(6, 9, 4)	0.716371	0.817392	0.810918	0.937289 , Monte Carlo
5(c)	(9, 21, 9)	0.654461	0.906606	0.904804	0.957183 , Monte Carlo

For Figure 5(a), the actual honest difference is negative:

$$p_{\text{exact}} - p_B = -0.007568.$$

For Figure 5(b), the actual difference is

$$0.094547,$$

which is below 10 percent. Figure 5(c) retains an honest difference of approximately 25.03 percent against the special decoder, but the optimized decoder using the same all-H data exceeds the honest ParitySolve score. None of these comparisons supplies a sound verification gap.

The Figure 5(a) optimized value is exact:

$$\frac{27863673495}{34359738368} \approx 0.8109396293.$$

The Figure 5(b) and 5(c) values are fixed-seed Monte Carlo estimates with Wilson intervals recorded in `results/figure5_corrected.csv`.

6. Corrected statistical uncertainty

Each complete repetition returns one correct-or-incorrect bit. Let Y be its correctness indicator. Irrespective of the internal decomposition into definitive branches and random guesses,

$$Y \sim \text{Bernoulli}(p).$$

For r independent repetitions, the empirical accuracy has standard error

$$\boxed{\text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{r}}}.$$

The expression used in the original paper and scripts,

$$\sqrt{\frac{1-p}{2r}},$$

accounts for the random final guess conditional on failure but omits the fluctuation in whether a definitive branch occurs.

At the exact Figure 5(b) value with $r = 1000$, the two formulas give

$$0.009723 \quad \text{and} \quad 0.012383,$$

respectively.

7. Effect on the conclusions

Retained

- DCP sample construction;
- post-Fourier reflection phase state;
- complementary-label collision mechanism;
- selected ParitySolve branch and exact parity output;
- published collision expressions as upper and lower bounds;
- circuit as a structured hardware-sensitive workload;
- original clean/noisy simulations as illustrations of their specified models.

Strengthened

- the special all-H outcome has probability $1/N$ for every n ;
- the associated decoder is Bayes-optimal for one complete all-H measurement record.

Corrected or withdrawn

- p_B is not a general soundness bound;

- $p > p_B$ does not verify the claimed quantum-computation capability;
- the statement that no analytic ideal probability is available is replaced by the exact formula above;
- Figure 5 parameter gaps must use exact honest probability rather than only an upper bound;
- the standard-error formula is replaced by the Bernoulli formula.

Surviving scientific classification

The original construction is best classified as a **DCP-derived circuit benchmark or workload**. Calling it an adversarial capability-verification protocol would require an explicit null class and a proved optimal soundness bound against that entire class. The 2022 threshold does not provide such a bound.

8. Separate 2026 follow-up

The repository also contains a later phase-twirled, heralded DCP construction under **witness/**. It is not a repair already present in the 2022 article. Its ideal theorem concerns a narrower capability: a nonseparable joint measurement across two trusted DCP cells, against adaptive separable instruments across the cell partition.

The follow-up is a DCP-specific maximum-confidence witness. It has an ideal confidence gap of 1 versus at most $3/4$, a sequential finite-sample theorem, and a four-qubit Clifford implementation for $N = 2$.

The 2022 article predates Lee and Bae's June 2026 explicit GLOBAL-versus-SEP maximum-confidence framework as an earlier trusted-input attempt to test joint processing, but it did not contain their sound theorem. The historical relationship and priority boundary are documented in **witness/HISTORICAL_CONTEXT.md**.

9. AI-assisted research and responsibility

This August 2026 correction and follow-up analysis were developed with substantial assistance from OpenAI's GPT-5.6 Pro. The model was used for literature discovery, adversarial mathematical checking, derivations, numerical-test design, code generation and review, notation consistency, and technical-document preparation.

The AI-assisted reanalysis helped identify the multi-sample failure of p_B , prove the one-sample optimality statement, derive and verify exact probability formulas, formulate the phase-twirled witness, and recognize its relationship to later maximum-confidence certification work.

GPT-5.6 Pro is not an author and cannot assume responsibility for the work. Ruge Lin reviewed the arguments, selected the claims included in the repository, and accepts responsibility for their accuracy. The executable tests, source code, machine-readable results, and stated limitations are provided for independent checking.

10. Reproducibility

The decisive calculations can be reproduced from a modern Python environment without Qibo:

```
python -m pip install -e .[test]
python examples/reproduce_counterexample.py
python examples/reproduce_one_sample_optimality.py
python examples/reproduce_figure5.py
python scripts/verify_historical_files.py
python scripts/validate_committed_results.py
pytest -q
```

The committed machine-readable outputs are:

- results/correction_core_results.json;
- results/one_sample_optimality.json;
- results/figure5_corrected.csv;
- results/heralded_witness_results.json;

- `results/validation_crosschecks.json`.

Exact rational, integer, and combinatorial quantities are validated exactly. Floating-point linear-algebra diagnostics are validated within the numerical tolerances encoded by the tests; portable byte-identical floating-point output is not claimed.

The six original scripts remain unchanged. They require their historical software stack and are not imported by the corrected tests.

Appendix A. Exact outcome table for the smallest counterexample

For $N = 4$, the integer counts $16q_s(r, y)$, with columns ordered by $(r = 0, y = 0, 1, 2, 3)$ followed by $(r = 1, y = 0, 1, 2, 3)$, are

s	0,0	0,1	0,2	0,3	1,0	1,1	1,2	1,3
0	4	4	4	4	0	0	0	0
1	4	0	2	2	0	4	2	2
2	4	4	0	0	0	0	4	4
3	4	0	2	2	0	4	2	2

The $y = 1$ columns reveal parity exactly. Conditional on $y \in \{2, 3\}$, even secrets force a fixed reflection result while odd secrets produce an unbiased result, creating the two-sample correlation.

Appendix B. Repository status

- Historical code snapshot: immutable tag and release `paper-2022-original`.
- Author correction branch: `author-correction-2026`.
- Original six Python scripts: preserved byte-for-byte.
- Current note: author-maintained, not an APS Erratum.
- Follow-up witness: separate, ideal-model, non-peer-reviewed.
- AI assistance: explicitly disclosed, with responsibility retained by Ruge Lin.